

1.Addtion matrix

```
import java.util.Scanner;

public class MatrixAddition {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");

        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");

        int cols = sc.nextInt();

        int[][] A = new int[rows][cols];

        int[][] B = new int[rows][cols];

        int[][] C = new int[rows][cols];

        System.out.println("Enter elements of First Matrix:");

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                A[i][j] = sc.nextInt();

            }

        }

        System.out.println("Enter elements of Second Matrix:");

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                B[i][j] = sc.nextInt();

            }

        }

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                C[i][j] = A[i][j] + B[i][j];

            }

        }

        System.out.println("Sum of Matrices:");

        for (int i = 0; i < rows; i++) {
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        for (int j = 0; j < cols; j++) {
            System.out.print(C[i][j] + " ");
        }
        System.out.println();
    }
    sc.close();
}
}

```

2.transpose matrix

```

import java.util.Scanner;

public class MatrixTranspose {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();
        System.out.print("Enter number of columns: ");
        int cols = sc.nextInt();
        int[][] matrix = new int[rows][cols];
        System.out.println("Enter matrix elements:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }
        System.out.println("Original Matrix:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                System.out.print(matrix[i][j] + " ");
            }
            System.out.println();
        }
    }
}

```

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        System.out.println("Transpose Matrix:");
        for (int i = 0; i < cols; i++) {
            for (int j = 0; j < rows; j++) {
                System.out.print(matrix[j][i] + " ");
            }
            System.out.println();
        }
        sc.close();
    }
}

```

3.spiral matrix

```

import java.util.Scanner;

class SpiralMatrix {

    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int r = sc.nextInt();
        int c = sc.nextInt();
        int a[][] = new int[r][c];
        for(int i = 0; i < r; i++) {
            for(int j = 0; j < c; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        int top = 0, bottom = r - 1;
        int left = 0, right = c - 1;
        while(top <= bottom && left <= right) {
            for(int i = left; i <= right; i++)
                System.out.print(a[top][i] + " ");
            top++;
            for(int i = top; i <= bottom; i++)
                System.out.print(a[i][right] + " ");

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right--;
if(top <= bottom) {
    for(int i = right; i >= left; i--)
        System.out.print(a[bottom][i] + " ");
    bottom--;
}
if(left <= right) {
    for(int i = bottom; i >= top; i--)
        System.out.print(a[i][left] + " ");
    left++;
}
}
}
}

```

4.saddle point

```

import java.util.Scanner;

class SaddlePoint {

    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int a[][] = new int[n][n];

        for(int i = 0; i < n; i++) {

            for(int j = 0; j < n; j++) {

                a[i][j] = sc.nextInt();

            }

        }

        boolean found = false;

        for(int i = 0; i < n; i++) {

            int min = a[i][0];

            int col = 0;

            for(int j = 1; j < n; j++) {

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        if(a[i][j] < min) {  
            min = a[i][j];  
            col = j;  
        }  
    }  
    boolean saddle = true;  
    for(int k = 0; k < n; k++) {  
        if(a[k][col] > min) {  
            saddle = false;  
            break;  
        }  
    }  
    if(saddle) {  
        System.out.println("Saddle Point = " + min);  
        found = true;  
        break;  
    }  
}  
if(!found)  
    System.out.println("No Saddle Point");  
}  
}
```